CSCE 4930: Hardware Architectures for AI and ML Research Paper Guidelines & Rubric

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The research paper is **NOT OPTIONAL**. The paper grade represents **20%** of the total course grade.

A **team of exactly 2 students** will write a survey-style research paper on a subject of their own choice, related to hardware architectures for AI and ML. The paper should present a **critical literature survey** of methods, architectures, numerical formats, and/or systems in the chosen topic, with exposure of applications as well as **analysis of strengths, shortcomings, and future directions of the surveyed work**. *You are also required to design and run a small experiment of your own that illustrates one of the points raised in your survey*, and to report on its setup, results, and what it demonstrates.

Guidelines

- The paper is intended to be completed in **pairs** (exactly 2 students per team).
- Choose a topic relevant to the course: a hardware technique, numerical format, accelerator architecture, or deployment paradigm for efficient ML training and/or inference.
- The paper should be written in **ACM two-column conference format** (e.g., using the `acmart` class with the `sigconf` option), and should be between **6 and 9 pages**, including references. You may use **Overleaf**, a local LaTeX installation, or any other tool you prefer.
- The final paper manuscript should be submitted on **Canvas** as a PDF.
- To check the originality of your paper, you should submit it through **Turnitin** (enrollment key: `A79E-Z91Y-DQNB-PMJB`). Your paper's overall similarity score should be kept low (as a guideline, below 20%); it is your team's responsibility to revise the paper if this threshold is exceeded, well ahead of the final deadline.

What the Paper Must Cover

These are the points that must be discussed in the paper, not the sections of the paper. For example, the topic and motivation will be mentioned in the abstract and the introduction, not as their own section. The survey and the analysis can be one section with multiple subsections, or more than one section.

1. **Topic and motivation:** A clear statement of the problem or design space you are surveying, and why it matters for efficient AI/ML training or inference.
2. **Literature survey (minimum 6 papers):** A survey of **at least six published papers** on your topic. For each, summarize the core idea, the architectural or algorithmic approach, and the key reported results. At least **three of the six papers should be from the last**

five years, to ensure the survey reflects the current state of the field. Course papers and readings may be included in your six, but at least half should come from your own literature search beyond the syllabus.

3. **Critical analysis:** For the surveyed work as a whole (not just paper-by-paper), discuss:
   - **Strengths**: what problems these approaches solve well, and under what conditions.
   - **Shortcomings**: limitations in accuracy, hardware cost, generality, scalability, or applicability.
   - **Future directions**: open problems or promising extensions, based on the gaps you identified.
4. **Experiment:** Design and run a small, well-scoped experiment that illustrates *one specific point* raised in your survey (e.g., an accuracy/efficiency trade-off, a claimed speedup, a resource cost, or a failure mode). This can use software simulation, FPGA emulation, an existing framework (e.g., hls4ml, FINN, PyTorch/TFLite quantization), or a simplified analytical/roofline-style model — the experiment should be clearly connected to a specific claim from the papers you surveyed.
5. **Conclusion and future work:** Synthesize what your survey and experiment together show, and what you would investigate next.

Grading Rubric

- **Literature survey (20%):** Coverage of a minimum of six published papers with accurate, substantive summaries of what each paper does and finds; appropriate mix of foundational and recent (last five years) work; the survey clearly sets up the analysis and experiment that follow.
- **Critical analysis (25%):** Depth and accuracy of the discussion of strengths, shortcomings, and future directions. This should synthesize across the surveyed papers rather than simply restating each paper's own conclusions, and should be well supported by evidence from the literature.
- **Experiment (25%):** A meaningful, well-designed experiment that clearly illustrates a specific point from the survey; sound methodology; results are presented and interpreted correctly, with a clear link back to the claim being illustrated.
- **Writing and formatting (15%):** Adherence to the ACM double-column format and page limit; correct in-text citations and a properly formatted reference list; clear organization (section structure, flow of ideas); the paper reads as a cohesive whole even though it was written by a team.
- **Conclusion and future work (15%):** A meaningful synthesis of what was learned from the literature survey and the experiment, with well-reasoned suggestions for future work.
- **Academic integrity (pass/fail gate):** Turnitin similarity should be kept low (guideline: below 20%); any use of AI tools must be acknowledged per the policy below. Violations result in a grade of zero on the paper, independent of the weighted score above.

Milestones

- **Topic and team selection — Monday, October 5, 2026:** Submit your team (2 members) and chosen topic on Canvas, along with a short (one-paragraph) description of the angle you plan to take and an initial list of candidate papers.
- **Draft due — Thursday, November 12, 2026:** Submit a draft of the paper including a substantially complete literature survey and a description (and, if possible, preliminary results) of your planned experiment.
- **Final paper due — Thursday, December 3, 2026:** Final submission of the complete paper (PDF) on Canvas, including the finished experiment and results. **There is no late policy for the paper**; no submissions will be accepted after the due date.

Can I Use AI Tools?

Anything you submit is your responsibility. If you use an AI tool at any point while preparing this paper (e.g., for background reading, brainstorming, LaTeX formatting, or grammar and style editing), you must clearly acknowledge this use in a short statement at the end of the paper. You must verify the correctness of anything an AI tool produces or suggests, including any citations, and be able to explain and defend every claim in your paper. **AI tools must not be used to write the paper's content for you**, and an AI tool **cannot** be cited as a reference. Submitting content generated by an AI tool as your own original work, or citing a source that does not exist, is a violation of academic integrity.